

What is temperature?

From sensation to definition

Physics · 3A · 8 September 2026

A room, four cubes

(slide 8–9)

The diagnostic ends here. Now: a room, four cubes and a question. No calculators, no formulas: just your hands and your intuition. For now, that is the whole experiment.

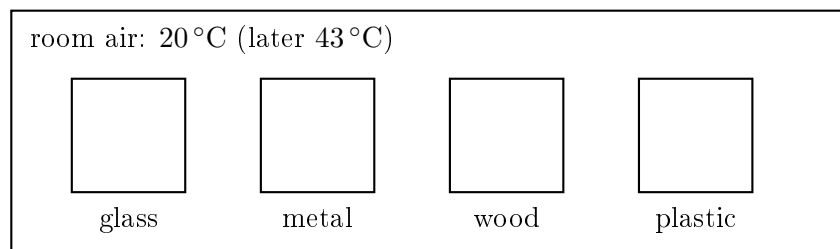


Figure 1: The four cubes in the room (schematic).

Before moving on. Four cubes, made of glass, metal, wood and plastic, sit in the same room at 20 °C. Which one feels coldest to the touch? A: glass. B: metal. C: wood. D: plastic. E: they are all the same. Go with your gut: this is not a trick question about physics you have not learned yet, it is about what your hand tells you.

The same question in a hot room

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Before moving on. Now the room is at 43 °C. The same four cubes. Which one feels coldest now? The options are the same, A to E.

What are their temperatures?

(slide 11)

Before moving on. Both times, all four cubes had been sitting in the room for hours. What are their temperatures, in the cold room and in the hot room? Think of a book and a spoon left overnight in a closed room: would you expect them to end up at different temperatures?

This question stays open: we come back to it at the end of the hour.

So, what is temperature?

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Let us find out, rigorously.

A first attempt: just define it

(slide 13)

Let us try this definition.

Temperature is the quantity measured by a thermometer.

Before moving on. Does this definition work?

And what is a thermometer? An instrument that measures temperature. Temperature → thermometer → temperature: the circle closes and we have defined nothing. This is called a *tautology*: a definition that quietly assumes what it was supposed to explain. To break out of it we need something that comes *before* the thermometer.

The same puzzle

(slide 14)

In the same room, at the same temperature, a tiled floor feels colder than a rug. It is the same trick as the cubes. For now we keep the question open: the answer is built over the next slides.

Before the thermometer

(slide 15)

You have seen it a thousand times. You sit on a cold chair: the chair warms up, you cool down, then it stops. You get into the swimming pool: freezing at first, fine ten minutes later. Water goes into the freezer and ends up at the freezer's temperature.

Before moving on. What do all these situations have in common?

Two bodies in contact. Something changes. Then it stops changing.

Thermal equilibrium

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A system is in **equilibrium** when its observable properties stop changing.

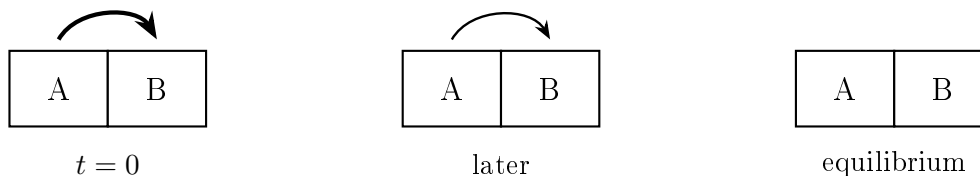


Figure 2: Two systems in contact. The arrow shows the energy passing from A to B: large at first, smaller later, zero at equilibrium.

Notice that we defined equilibrium without ever using the word *temperature*. “Being in thermal equilibrium with” is a *relation*: given two systems, it answers yes or no. (Definition after Tarsitani, 2005.)

Three properties, one not obvious

(slide 17)

This relation has three properties.

- **Reflexive**: every system is in equilibrium with itself. Obvious.
- **Symmetric**: if A is in equilibrium with B, then B is in equilibrium with A. Obvious.
- **Transitive**: if A is in equilibrium with C and B is in equilibrium with C, then A and B are in equilibrium with each other. Not obvious.

In symbols: $\forall a : a \sim a \quad a \sim b \Rightarrow b \sim a \quad a \sim b \wedge b \sim c \Rightarrow a \sim c$.

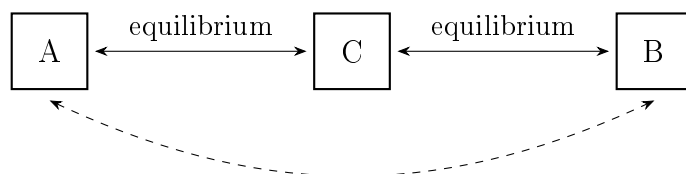
The transitive property has to be checked. With three cups of water at the same temperature you are not surprised that, if A matches C and B matches C, then A and B match too; but you have to actually check it, because nothing forces it to be true. A relation with all three properties has a name in mathematics: we will get to it shortly.

The Zeroth Law

(slide 18)

The transitive property has a name in physics: the **Zeroth Law of Thermodynamics**.

$$T(A) = T(C) \text{ and } T(B) = T(C) \implies T(A) = T(B)$$



$\Rightarrow$ A and B are in equilibrium with each other (Zeroth Law)

Figure 3: The Zeroth Law: if A is in equilibrium with C and B is in equilibrium with C, then A and B are in equilibrium with each other.

In practice: the thermometer reads 20°C on the water and 20°C on the metal, so water and metal are in equilibrium with each other, without ever putting them in contact. It is an experimental fact, not a logical necessity: it could have gone otherwise. Without it, a thermometer would only ever measure itself.

Why is it called “zeroth”?

(slide 19)

The First and Second Laws were established and numbered in the 1850s. Decades later, physicists noticed that both quietly assumed this more basic fact. It had to come first, but “First” and “Second” were taken. The only number left before one was zero: Ralph Fowler, 1930s. Physics realised, eighty years late, that it had taken the most basic thing for granted.

The order of discovery and the order of logic are not the same thing.

A little mathematics: an equivalence relation

(slide 20)

The three properties together define an **equivalence relation**. It behaves like “being equal”: not identical in everything, but equal in one specific respect. And it automatically divides its set into classes that never overlap.

An example: having the same year of birth. I was born in my own year: reflexive. If I share your year, you share mine: symmetric. If you and I were born in 2010, and you and Marco were born in 2010, then Marco and I were too: transitive.

A counterexample: “taller than”. I am not taller than myself: reflexivity fails. If I am taller than you, you are not taller than me: symmetry fails. It is an order relation: it puts things in a line, it does not group them.

A worked example: rational numbers

(slide 21)

A fraction is a *pair* of integers: $(1, 2)$ and $(2, 4)$ are different pairs. We regard them as equivalent when

$$(a, b) \sim (c, d) \iff a \cdot d = b \cdot c.$$

Check: $(1, 2) \sim (2, 4)$ because $1 \cdot 4 = 2 \cdot 2$. So $1/2$, $2/4$, $3/6$ are one point with three names.

A rational number is not a pair: it is the *whole class* $\{(1, 2), (2, 4), (3, 6), (50, 100), \dots\}$. Writing “ $1/2$ ” means naming the class through a convenient representative.

Equivalence classes are everywhere

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- **Days of the week:** 100 days from now is the same day of the week as 2 days from now, because 100 and 2 leave the same remainder when divided by 7. Seven classes.
- **The clock face:** 15:00 and 3 o'clock are the same class. A clock is a quotient set made visible.
- **Blood groups:** “same group as” divides everyone into A, B, AB, O. Four labels, millions of people.
- **Direction of a line:** “parallel to” is an equivalence relation. A direction is not a line: it is a class of lines.
- **Musical pitch class:** a note and the same note an octave higher have the same name. “C” is a class, not a single frequency.
- **Colour:** “red” is not an object. It is the name of a class of objects.

The definition, at last

(slide 23)

“Being in thermal equilibrium with” is an equivalence relation: reflexive and symmetric in an obvious way, transitive by the Zeroth Law. So it divides every thermodynamic system in the universe into classes.

Temperature is the name of the class.

A rational number is a class of fractions. A direction is a class of lines. A temperature is a class of systems.

Beyond the syllabus, for the record: $\tau : S/\sim \rightarrow \mathbb{R}$. Temperature is a function on the set of classes and assigns a different number to each class.

And the thermometer?

(slide 24)

Before moving on. Does a thermometer measure the object's temperature, or its own?

The circle is broken

(slide 25)

The answer is: both, provided equilibrium has been reached. A thermometer is a system with one property that is influenced by its surroundings and easy to read: the volume of a liquid, the pressure of a gas, the resistance of a conductor. You put it in contact: it joins the same class as the object and makes the label readable. Temperature was defined *before* the thermometer, not by it, and that is what breaks the tautology. It is also why a thermometer must be left in place before you read it: you are waiting for equilibrium.

A real problem, in one sentence. Two thermometers, one mercury and one alcohol, calibrated to agree at 0°C and 100°C, read a few tenths of a degree apart in between, because mercury and alcohol expand according to different laws. They agree on *which* systems are equivalent; they disagree only on the number in between. It is a scale problem, not a physics problem.

A check, with no new facts

(slide 26)

Before moving on. How long should you leave a thermometer in place? Why can you not read it immediately?

You are not reading the object: you are waiting for the thermometer to join the object's equivalence class, that is, to reach equilibrium with it. If you read it too early it has not yet "voted": its number still belongs to its own earlier class, not to the object's.

Temperature of what?

(slide 27)

A **thermodynamic system** is a system for which it makes sense to define a temperature: a macroscopic system with an "inside" that the surroundings can influence.

Careful with thermostats. Some systems hold their own temperature steady against the surroundings: the human body, a cat, a working fridge. They do have a temperature: they simply fight equilibrium instead of drifting into it.

Careful also with the opposite end. A single atom has no temperature. Neither does an idea or a radio wave. Temperature needs a system with many parts: we will see why on 2 October.

One list, three questions

(slide 28–29)

Here is the list: battery, magnet, yoghurt, block of wax, cat, tree, spring, resistor, bacterial colony, fridge, piece of ice, helium cylinder, boiling water, hot milk flask, the Moon.

Before moving on. Pass 1. For which of these does it make sense to give *one* temperature? (Not everything is uniform inside.)

Pass 2. Which ones are heading toward equilibrium with the room, and which are actively resisting it?

Pass 3. Which could be a thermometer? Battery, magnet, spring and resistor each have a property that changes *with* temperature: voltage, magnetisation, elasticity, resistance. Which property would you read? Would it be easy to measure?

Same list: look at it again for each question, do not just sort it once. Then, on your own: give four more examples of your own, taken from ordinary objects around you, and say which pass they belong to. (Exercise from Vicentini Missoni, 1989.) The answers are in the solutions at the end.

Heat and temperature

(slide 30)

Before moving on. Regarding heat and temperature, we can say that: A, they are equivalent concepts. B, a body receiving energy as heat must necessarily change temperature. C, a body receiving energy as heat may change temperature. D, temperature indicates the amount of heat a body contains. E, none of the above. (Quiz from Vicentini Missoni, 1989.)

The correct answer is C: “may”, not “must”. We come back to it with changes of state (20 October), where a body absorbs energy without changing temperature. D is the idea of heat as a substance that a body “contains”: a leftover of everyday language, as in “the heat it gives off”.

Closing the hook: the four cubes

(slide 31)

All four cubes were at the *same* temperature both times: the temperature of the room, first 20°C, then 43°C. Your hand does not measure temperature: it measures *how fast* energy leaves your finger.

Why does metal feel coldest? Metal conducts energy away from your skin fastest. Wood and plastic conduct slowly, so less energy leaves you per second and they feel closer to your own temperature.

And at 43°C? Above your skin’s temperature the flow reverses: energy now moves into your finger. Metal still moves it fastest, so now it should feel hottest.

Why does the rate depend on the material? This question stays open: 30 October.

Three things to take away

(slide 32)

1. **Sensation is not measurement.** Your hand reports a rate of energy flow, not a temperature.
2. “Being in thermal equilibrium with” is an equivalence relation. Its transitivity is the Zeroth Law, an experimental fact.
3. Temperature is the label of an equivalence class, like a rational number, like a direction. Defined before any thermometer exists.

Questions to keep in your pocket

(slide 33)

Not for today.

- **15 September, expansion:** the Eiffel Tower is taller in summer. By how much? Predict first.
- **6 October, heat:** is the Sun’s radiant heat actually heat? Does cold exist, or only less heat?
- **20 October, changes of state:** can a body receive heat without changing temperature?
- **30 October, conduction:** why is 20°C in air comfortable and 20°C in water freezing? Why does blowing on soup cool it, but blowing on your hands warms them?
- **No date, maybe no answer:** hot water and cold water into the freezer, which freezes first? The Mpemba effect is genuinely disputed: we will hold it open, honestly, not resolve it for a cheap answer.

Homework

(slide 34)

To hand in. First: finish the three-pass classification from today. Second: add four examples of your own, taken from ordinary objects around you. Everyone’s four will be different: that is the point, it is our first real data set. Short, no calculation. Next lesson, 11 September: how do you actually build a thermometer?

Solutions to the exercises

(slide 28–29)

The list in three passes.

- Pass 1.** One temperature makes sense for everything except the **Moon**: the lit side is above 100°C and the dark side is below -170°C , so it has no single temperature. The tree is a borderline case worth discussing (large and non-uniform, but usually accepted).
- Pass 2.** The ones fighting equilibrium, holding their own temperature, are the **cat**, the **bacterial colony** and a running **fridge**. Everything else drifts toward room temperature.
- Pass 3.** Thermometer candidates: the **battery** (read the terminal voltage), the **spring** (its length under a fixed load), the **resistor** (its resistance, with a multimeter). The **magnet** is a borderline case: a ferromagnet's magnetisation varies with temperature and collapses above the Curie temperature, so it is a poor thermometer, but proposing it is not wrong. It is a reminder that “readable” is a requirement, not just “sensitive”.

The four cubes (slide 11). The same temperature (option E) in both cases: the cubes have been in the room for hours, in equilibrium with the air and therefore with each other.